

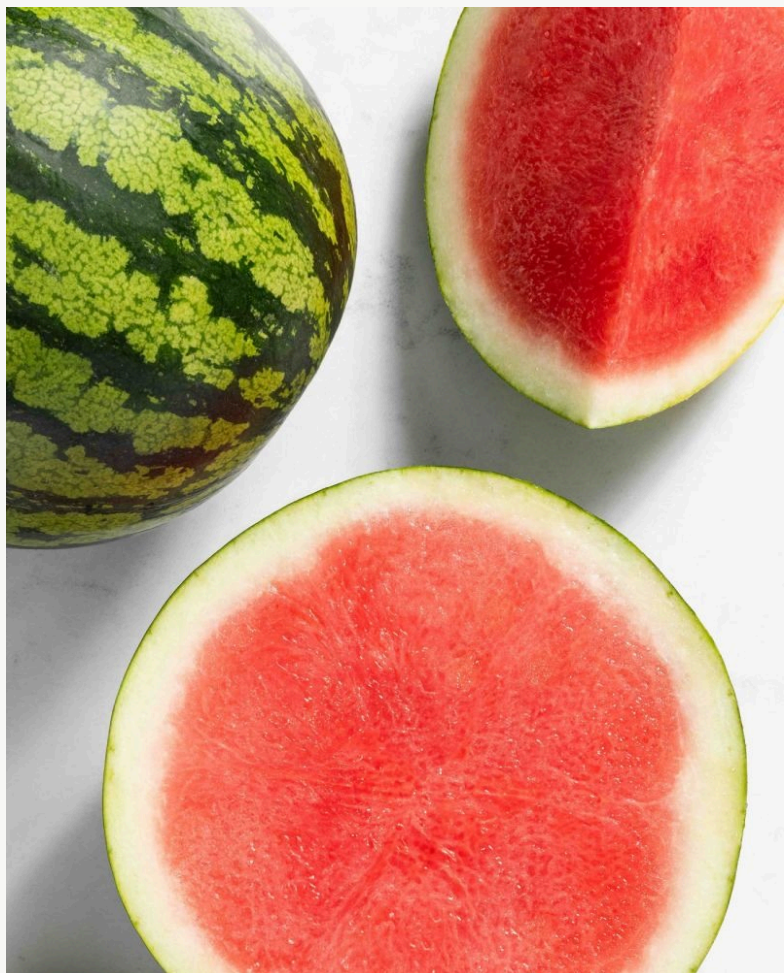
The 2026 Watermelon Internal Quality Benchmark

What could reducing watermelon shipment rejections from 3.5% to 1% be worth?

A field-based reference for U.S. watermelon growers, packers, and shippers — with a modeled savings scenario for a representative 60-million-pound seasonal operation.

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A shipment judged by a handful of fruit

A buyer may inspect and cut only a handful of fruit before deciding whether a shipment meets expectations. At the packing line, destructive sampling and the know-how of a few experienced tappers are the only line of defense. **Most of the fruit remains unseen inside.**

3.5%

PRACTICAL FIELD
REFERENCE

1%

TARGETED REJECTION
RATE

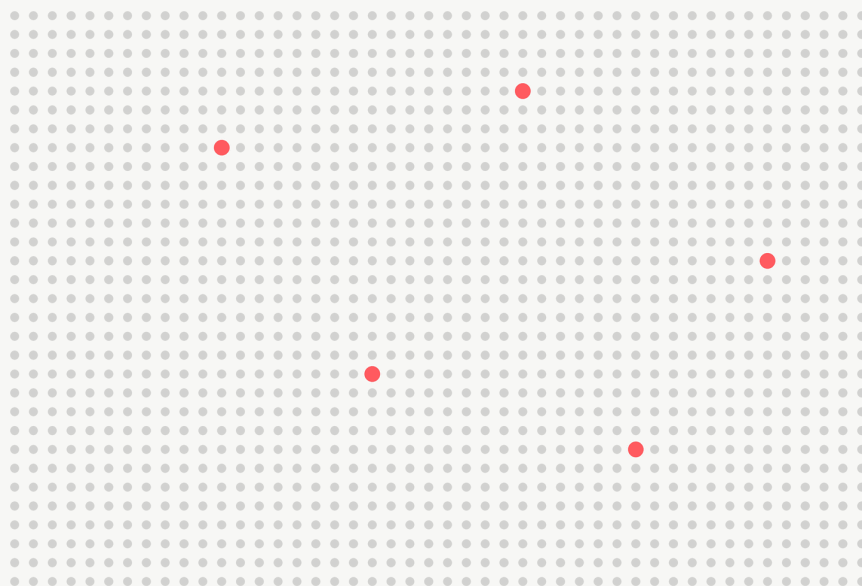
\$700K

UP TO — MODELED
ANNUAL SAVINGS, 60M
LB

40K lb

AVERAGE SHIPMENT SIZE

The 3.5% figure is a practical field reference, **not a statistically representative national average.** Strong producers may already perform materially better — and should be able to demonstrate it.



ILLUSTRATIVE. ONE SHIPMENT HOLDS THOUSANDS OF FRUIT; THE DECISION MAY REST ON A HANDFUL.

The sampling gap

Sampling is necessary — cutting every watermelon open is impossible. But internal conditions may not be distributed evenly, the result depends on which fruit are selected, and sampled fruit can no longer be sold as packed product.

The largest cost is not the fruit destroyed. It's the commercial decision made with incomplete information.

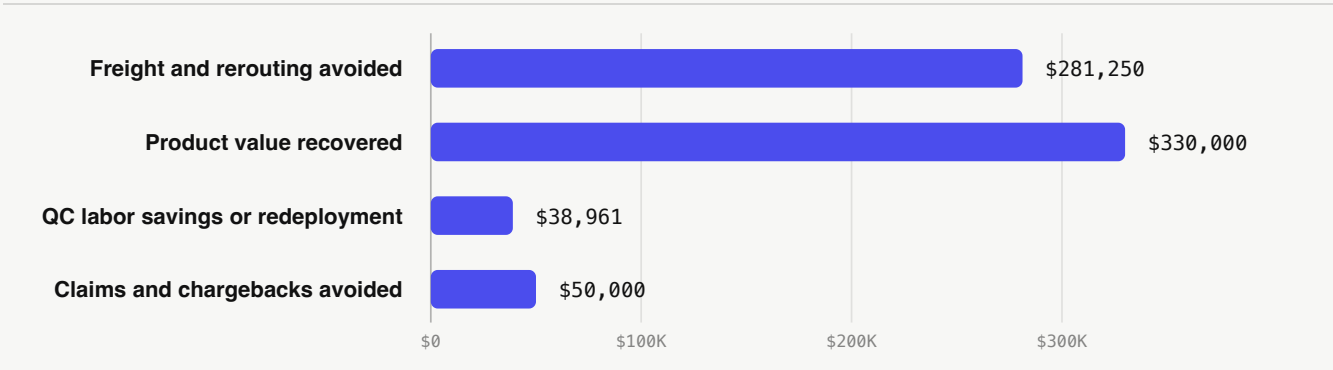
● SAMPLED ● UNSEEN

What the reduction is worth

For a representative 60-million-pound operation, the modeled savings of moving from a 3.5% rejection rate to a 1% target **can reach approximately \$700,000 per season** — from avoided freight and rerouting, recovered product value, QC labor savings or redeployment, and supported claims and chargebacks.

This is an upper-bound scenario using full capture of the modeled savings. It is a decision-making reference, not a guaranteed result. Actual savings require operation-specific validation.

SEASON VOLUME	REFERENCE SHIPMENT	SHIPMENTS PER SEASON	REJECTION RATE	REJECTIONS AVOIDED
60M lb	40,000 lb	1,500	3.5% → 1%	37.5



Total modeled annual savings \$700,211

\$700,000

UP TO — MODELED ANNUAL SAVINGS

Component values are exact model outputs and total \$700,211; the headline is rounded down. Actual savings depend on volume, rejection frequency, freight exposure, product value, labor configuration, claim history, and the share of rejection events that can be prevented.

The \$700,000 scenario excludes destructive-sampling savings, premium pricing, incremental revenue, Orbem pricing, and unverified consequences such as shelf-life deterioration, replacement obligations, disposal loss, or future buyer allocation. It is a model, not independently validated industry research.

Where the savings come from

Five places to put a number against your own operation. Four are modeled above; the fifth is operation-specific. The answer should be your own figure — not a universal promise.

01

Freight and rerouting avoided

Freight already spent on a rejected load, plus the cost of moving it somewhere else.

02

Product value recovered

Load value preserved when internal quality is known before the fruit ships.

03

QC labor savings or redeployment

Shift time currently absorbed by sampling and subjective internal-quality calls.

04

Destructively sampled fruit

Fruit cut open in a season, at packed value. Operation-specific, so excluded from the \$700,000.

05

Claims and chargebacks avoided

Documented claims tied to internal quality, net of value already counted under 02.

START WITH TWO NUMBERS — THE FIVE SOURCES ABOVE ASK FOR THE REST

01

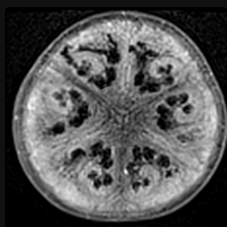
Volume and loads

Annual pounds, shipment count, and average payload.

02

Rejection incidence and reasons

Current rate, by buyer and destination — and what drove each event.



FROM A FEW SAMPLES TO EVERY-FRUIT INFORMATION

Orbem's Genus Echo inspects internal watermelon quality non-destructively at up to 44,000 lb/hour.

The difference is coverage. Sampling uses a few fruit to represent a shipment; every-fruit inspection can provide internal-quality information before fruit reaches the buyer — informing grading, holding, releasing, routing, field and variety analysis, and buyer-specific quality management.

BENCHMARK YOUR OPERATION

REQUEST A CONFIDENTIAL INTERNAL-QUALITY REVIEW

BENCHMARK
CONFIDENTIAL REVIEW

amkappes-cmd.github.io/orbem-watermelon
calendar.app.google/1pP3A1nvstA7CGHr8

METHODOLOGY AND LIMITATIONS — The \$700,211 scenario models a representative 60-million-pound seasonal operation moving from a 3.5% rejection rate to a 1% target, with capture set to 100%. It is an upper-bound thought exercise built on anonymized work with two U.S. watermelon producers, site observations, and broader field conversations — not a statistically representative

national survey and not independently validated industry research. It does not assume that every rejection is caused by internal quality or that every affected shipment can be prevented. Actual savings vary with volume, rejection frequency, freight exposure, product value, labor configuration, claim history, and the share of rejection events that can be prevented.